Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal line” and “reflection” to explain your model or result. Sentence starters are optional.

normal line: a reference line perpendicular to a surface

reflection: a wave returning from a boundary

virtual image: an image at an apparent origin of rays, not their meeting point

01 Bring back an earlier idea

Sensors 1.02 m apart record delay 0.003 s.
Find wave speed. If 0.0005 s is added to the delay, find the incorrect speed and describe the bias.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; measure angles from the normal. The image is virtual.

Worked reasoning: An object 2 m in front of an ideal plane mirror has a virtual image 2 m behind it. The object-image separation is 4 m; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

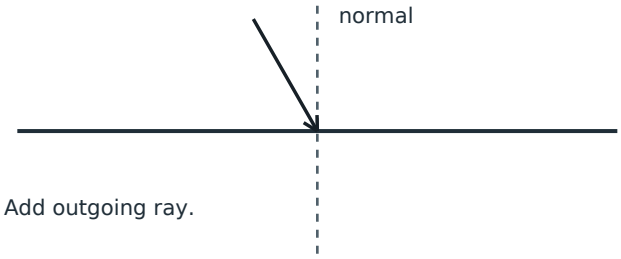
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw the normal at the mirror. Complete a reflected ray for an incident ray 30° from normal.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

04 Solve and justify

A ray is 20° from the mirror surface.
Find incidence angle from normal and angle between incident and reflected rays.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why is the mirror image called virtual rather than a real object behind the glass?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

Echoes return at 0.004 s and 0.006 s with the same wave speed.
Compare ranges. Does delay alone determine mass?

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

An object moves .50 m toward a fixed mirror.
Find image movement and change in object-image separation.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

An object moves toward a fixed plane mirror at 1.2 m/s.
Find image velocity relative to the mirror and the rate at which object-image separation decreases.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3